ADIL ARYA

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EDUCATION

Columbia University — **New York City, New York**
Master of Science in Artificial Intelligence — Sep 2026 – Dec 2027

University of Minnesota - Twin Cities — **Minneapolis, Minnesota**
Bachelor of Science in Computer Science — Sep 2024 – May 2026

- **Honors**: 3.912 / 4.0 GPA, 5x CSE Dean's List, High Honors
- **Courses**: Algorithms & Data Structures, Operating Systems, Machine Learning, Computer Vision, Robotics, NLP, Artificial Intelligence I & II
- B.S. completed in 2 years (90+ credits, 8 graduate-level courses).

WORK EXPERIENCE

Minnesota Robotics Institute — **Minneapolis, Minnesota**
Undergraduate Researcher — Oct 2024 – May 2026

- Built a 3D-reconstruction evaluation pipeline benchmarking **GLOMAP vs COLMAP** across 30 trials (Chamfer, precision/recall, F-score); raised F-score **0.79 to 0.89** and cut Chamfer error **~30%**, driving the lab's adoption of GLOMAP as its reconstruction backbone (Python, NumPy, Pandas).
- Architected **CableMind**, an autonomous robotic system pairing a modular **state machine** (perception to planning to execution via xArm SDK) with a **LoRA-fine-tuned OpenVLA (Llama-2-7B)** policy mapping language instructions to robot trajectories; **~100%** success on single-clip, **~60%** on randomized 4-clip routing.

Divertica Inc. — **Remote**
Computer Vision Intern — Jun 2025 – Aug 2025

- Shipped an end-to-end **ML pipeline** processing **4K/60fps** video into structured retail planograms: frame stitching, object detection, and OCR stages wired into one system.
- Fine-tuned and evaluated detection models (**YOLOv8/v12** on **SKU110K**, ResNet50, OCR) to **0.87 F1 / 0.90 mAP50**; found the obvious upgrade (v12) gave no real gain, diagnosed a **~630K-class** scaling bottleneck, and proposed a **retrieval-based** design.

PROJECTS

TrialFind
Origin House O1 Summit Hackathon — Apr 2026 – Apr 2026

- **1st Place, TinyFish Sponsor Track** (24-hour build, team of 3): agentic system matching cancer patients to clinical trials, pairing an **LLM summarization agent** with **parallel browser agents** searching live trial registries.
- Orchestrated **6 concurrent browser agents (up from 2)** across 3 API providers, parallelizing real-time multi-website search across trial registries.

Physical IDE
Google I/O Build with AI Hackathon x Google Cloud Labs — May 2026 – May 2026

- **Finalist, top 7 of 50+ teams** (12-hour build, team of 3): real-time hardware assembly verification agent streaming webcam frames to **Gemini Live API**, collapsing **3 model calls (vision, reasoning, speech) into one round-trip.**
- Owned the serving backend: a **FastAPI WebSocket gateway and state machine on Google Cloud Run Gen2 with session affinity**, gating inference via **MediaPipe Hands** so the model evaluates only when the user's hands clear the frame (Python, WebSockets, Docker).

Recursive Ray Tracer
CSCI 5607, University of Minnesota — Jan 2026 – Apr 2026

- Implemented a recursive ray tracer in **C++** – ray-primitive intersection, Blinn-Phong shading, reflection/refraction via Schlick's approximation, and Monte Carlo diffuse interreflection.

RESEARCH

Imitation of Imitations: Model Collapse under Synthetic Training Data — 2026
M. Taneja, A. Arya
Controlled study of model collapse: pretrained GPT-2 on clean/synthetic code mixtures and evaluated on rare languages (Rust, CUDA, Assembly). Performance degrades gradually up to 90% synthetic data, then collapses sharply (3× perplexity) at 100% — identifying even a small fraction of human data as a critical grounding signal.

SKILLS

Programming Languages: Python, C++, SQL, JavaScript / TypeScript
ML & Deep Learning: PyTorch, JAX, Hugging Face Transformers, LoRA fine-tuning, scikit-learn, NumPy, Pandas
ML Infrastructure & Serving: Docker, Google Cloud Run, FastAPI, WebSockets, REST APIs, model serving, Git, Linux
NLP & LLMs: LangChain, OpenAI / Anthropic APIs, Agentic systems, synthetic data generation
Computer Vision: OpenCV, torchvision, Ultralytics YOLO, Structure-from-Motion